

Demonstration of Proposed Features

Date	08.10.2026
Team ID	UNM1611BHAVANA
Project Name	AI FitTrack API
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	User Registration	Creates a new user account (name, email, password) via POST /api/auth/register.	Implemented	Yes	Returned 201 Created and a JWT token.
2	User Login (JWT Authentication)	Authenticates a user with email and password via POST /api/auth/login and issues a JWT token.	Implemented	Yes	Returned 200 with token and user id.
3	User Profile (Protected Route)	Returns the logged-in user's profile via GET /api/auth/profile using the token.	Implemented	Yes	Returned 200 in about 12 ms.
4	AI Workout Recommendation	Generates personalised workout suggestions through the Gemini AI service via POST /api/ai/workout-recommendation.	Implemented	Yes	Returned 200; response time about 7.8 s.
5	AI Fitness Insights	Analyses totalWorkouts, averageDuration and totalCaloriesBurned to produce fitness insights via POST /api/ai/fitness-insights.	Implemented	Yes	Input validation shown: 400 Bad Request when fields are missing or invalid.
6	Workout Management	Workout controller and routes for managing user workout records (workoutController.js, workoutRoutes.js).	Implemented	No	Present in code and Postman collection; not shown in the attached screenshots.

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	5
Overall Implementation Rate (%)	100% (6 of 6 implemented)

Demonstration Evidence:

Figure 1 – Authentication demo. The server starts with npm start (Server running in development mode on port 5000, MongoDB Connected: localhost). POST /api/auth/register returns 201 Created and a JWT token with the user id.

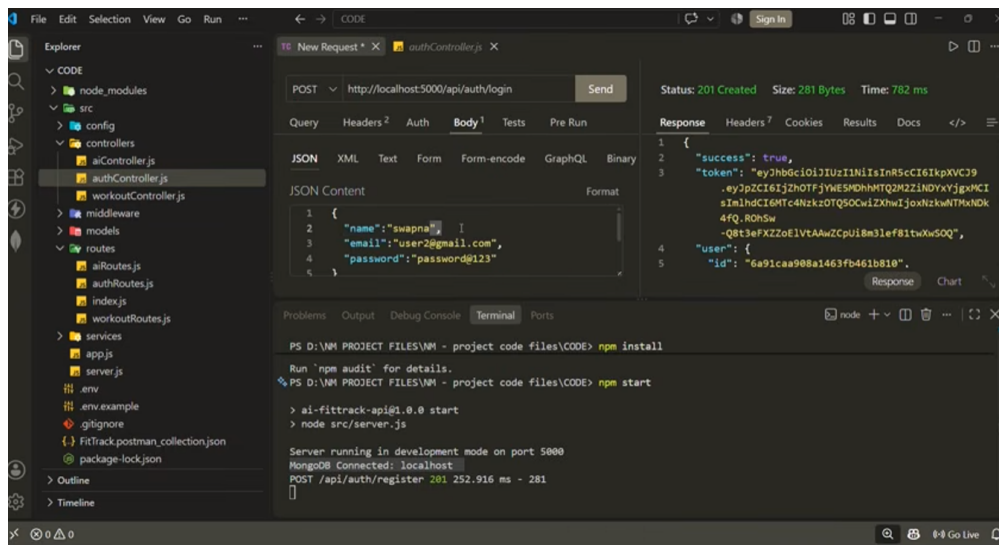


Figure 1: Auth API tested in the API client with the VS Code terminal showing the running server.

Figure 2 – AI endpoints demo. The server log shows register (201), login (200), profile (200) and workout-recommendation (200). The fitness-insights request returned 400 Bad Request with the message asking for totalWorkouts, averageDuration and totalCaloriesBurned, demonstrating input validation.

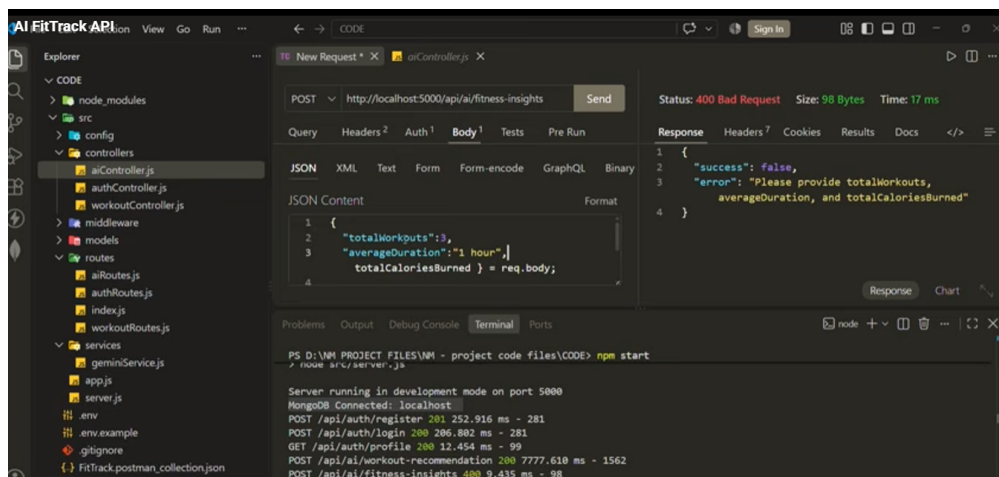


Figure 2: POST /api/ai/fitness-insights validation response and server request log.